

Quantum Systems Analysis

Finite and Infinite Potential Wells: Dynamics, Eigenstates, and Numerical Simulation

Sofía Carolina Quintanilla Galindo

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Infinite Potential Well: Dynamical Analysis

Quantum Revivals

Analytical derivation of the revival time T .

The problem asks for the time after which a quantum particle returns to its exact initial state — the same wavefunction shape and phase. This phenomenon is known as a *quantum revival*.

Any quantum state can be decomposed into a superposition of stationary modes — the eigenstates of the system — each oscillating at a frequency associated with an energy E_n . Starting from the general wavefunction in an infinite potential well,

$$\psi(x, t) = \sum_{n=1}^{\infty} c_n \phi_n(x) e^{-iE_n t/\hbar},$$

with eigenenergies

$$E_n = \frac{n^2 \pi^2 \hbar^2}{8ma^2},$$

where the factor of 8 in the denominator corresponds to a well of width $L = 2a$ (compared to the convention $L = a$ in Griffiths, which yields a factor of 2).

We require $\psi(x, t + T) = \psi(x, t)$, which means that all phases must complete an integer number of full cycles:

$$e^{-iE_n T/\hbar} = 1 \quad \forall n.$$

This holds when $E_n T/\hbar = 2\pi k_n$ for some $k_n \in \mathbb{Z}$. Evaluating the time-evolved wavefunction,

$$\psi(x, t + T) = \sum_{n=1}^{\infty} c_n \phi_n(x) e^{-iE_n t/\hbar} e^{-iE_n T/\hbar},$$

and substituting the eigenenergies, the condition becomes

$$\frac{E_n T}{\hbar} = \frac{n^2 \pi^2 \hbar T}{8ma^2} = 2\pi n^2,$$

which gives

$$T = \frac{16ma^2}{\pi \hbar}.$$

Substituting $a = L/2$, the revival time is

$$T = \frac{16m(L/2)^2}{\pi \hbar} = \frac{16mL^2}{4\pi \hbar} = \boxed{\frac{4mL^2}{\pi \hbar}}.$$

Computational Simulation

2.1 Case a: Superposition of Four Eigenstates

Normalization constant A . The squared modulus $|\psi(x, t)|^2$ represents the probability density of finding the particle at position x at time t . The total probability must equal unity, which imposes the normalization condition

$$\int_{-a}^a |\psi(x, t)|^2 dx = 1.$$

The initial wavefunction is a linear superposition of four stationary eigenstates $\phi_n(x)$ with complex coefficients, scaled by a normalization constant A :

$$\psi(x, 0) = A [\phi_1(x) + 0.7 \phi_2(x) - i 1.2 \phi_3(x) + (2 - i) \phi_4(x)].$$

To normalize, we write $\psi(x, 0) = A \sum_{n=1}^4 c_n \phi_n(x)$ and use the orthonormality of the eigenstates,

$$\int_{-a}^a \phi_n(x) \phi_m(x) dx = \delta_{nm},$$

to evaluate

$$\int_{-a}^a |\psi(x, 0)|^2 dx = |A|^2 \sum_{n=1}^4 |c_n|^2.$$

Computing each term,

$$|c_1|^2 = 1, \quad |c_2|^2 = 0.49, \quad |c_3|^2 = 1.44, \quad |c_4|^2 = 5,$$

so $\sum |c_n|^2 = 7.93$. Setting the integral to unity yields

$$A = \frac{1}{\sqrt{7.93}}.$$

Two-dimensional probability density $|\psi(x, t)|^2$. To generate this plot, the wavefunction is evaluated at each point of a spatial and temporal grid using the eigenstate expansion. Each term evolves with a complex temporal phase associated with its energy level. The squared modulus gives the probability density, which is stored in a matrix and displayed as a heatmap where color encodes probability intensity.

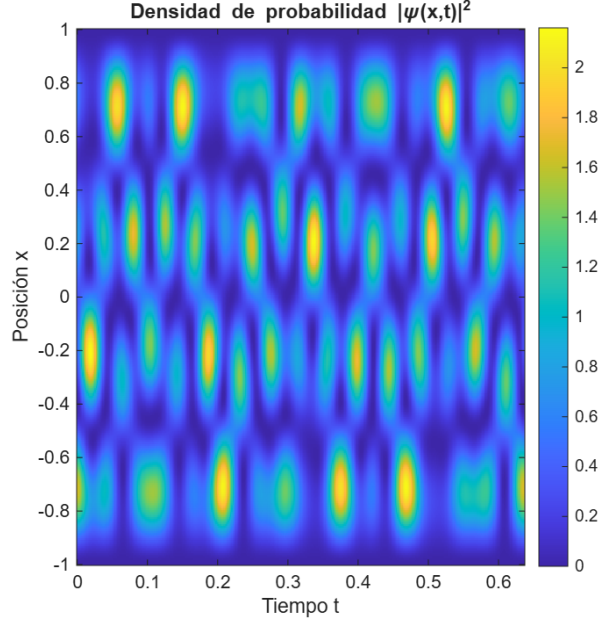


Figure 1: Probability density $|\psi(x,t)|^2$ for a superposition of four eigenstates.

The heatmap reveals interference patterns, wave oscillations, and the revival phenomenon: the wavefunction recovers its initial shape at $t = T$, producing the periodic pattern visible in the figure. Warmer regions correspond to higher probability of finding the particle.

Expected position $\langle x \rangle(t)$. Using the probability density computed above, the expectation value of position is calculated at each time step via numerical integration:

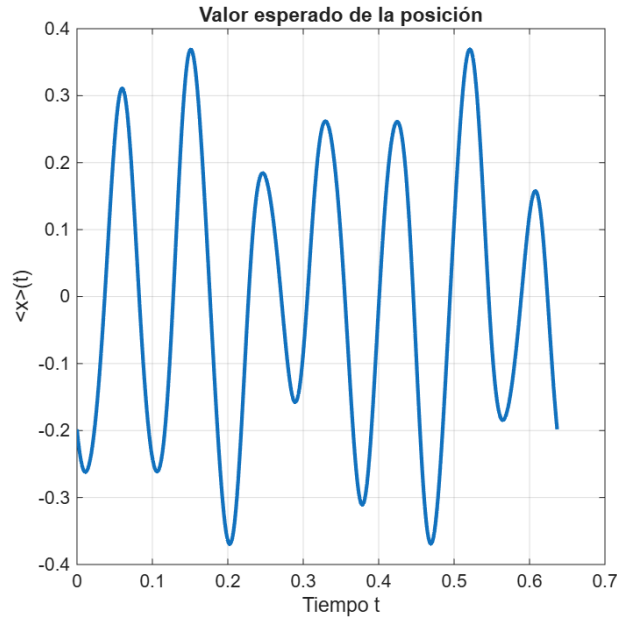


Figure 2: Expected position $\langle x \rangle(t)$ for a superposition of four eigenstates.

The curve represents the probability-weighted average position of the particle over time — analogous to a center of mass of the probability distribution. Since the initial state combines eigenstates with complex, asymmetric phases, the expectation value oscillates in a non-trivial, non-periodic manner reflecting the beating between the different energy frequencies involved.

2.2 Case b: Gaussian Pulse

Normalization constant A . The initial wavefunction is a Gaussian pulse centered at position $x = c$ within the well:

$$\psi(x, 0) = A \exp\left(-\frac{(x - c)^2}{b^2}\right),$$

where b controls the pulse width and c its center. Applying the normalization condition,

$$A^2 \int_{-a}^a \exp\left(-\frac{2(x - c)^2}{b^2}\right) dx = 1.$$

When b is small, the pulse is well-localized within $[-a, a]$ and the integration limits can be extended to $\pm\infty$:

$$\int_{-\infty}^{\infty} \exp\left(-\frac{2(x - c)^2}{b^2}\right) dx = \sqrt{\frac{\pi b^2}{2}}.$$

Solving for A ,

$$A = \left(\frac{2}{\pi b^2}\right)^{1/4}.$$

Two-dimensional probability density $|\psi(x, t)|^2$. The initial Gaussian pulse is expanded as a linear combination of the infinite well eigenstates $\phi_n(x)$, with coefficients c_n computed via numerical inner products. The wavefunction at time t is then reconstructed as

$$\psi(x, t) = \sum_n c_n \phi_n(x) e^{-iE_n t/\hbar},$$

and the probability density $|\psi(x, t)|^2$ is stored in a spatiotemporal grid and visualized as a heatmap.

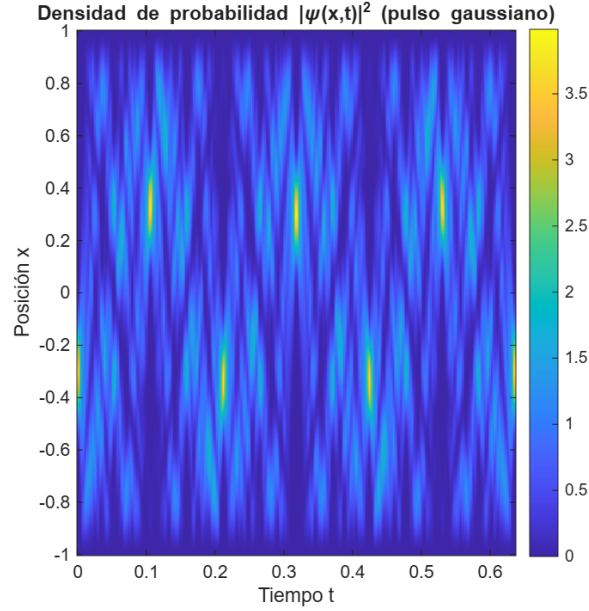


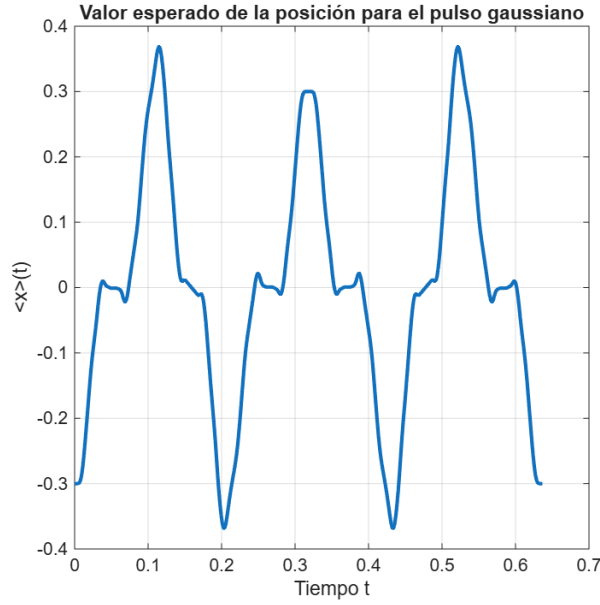
Figure 3: Probability density $|\psi(x, t)|^2$ for a Gaussian pulse.

The pulse is initially localized near $x = -0.3$. Over time, the wavefunction disperses and reflects off the well walls, generating interference patterns. The vertical bright bands near $x \approx \pm 0.3$ correspond to partial revivals, where constructive interference between energy modes reconstructs the initial probability concentration.

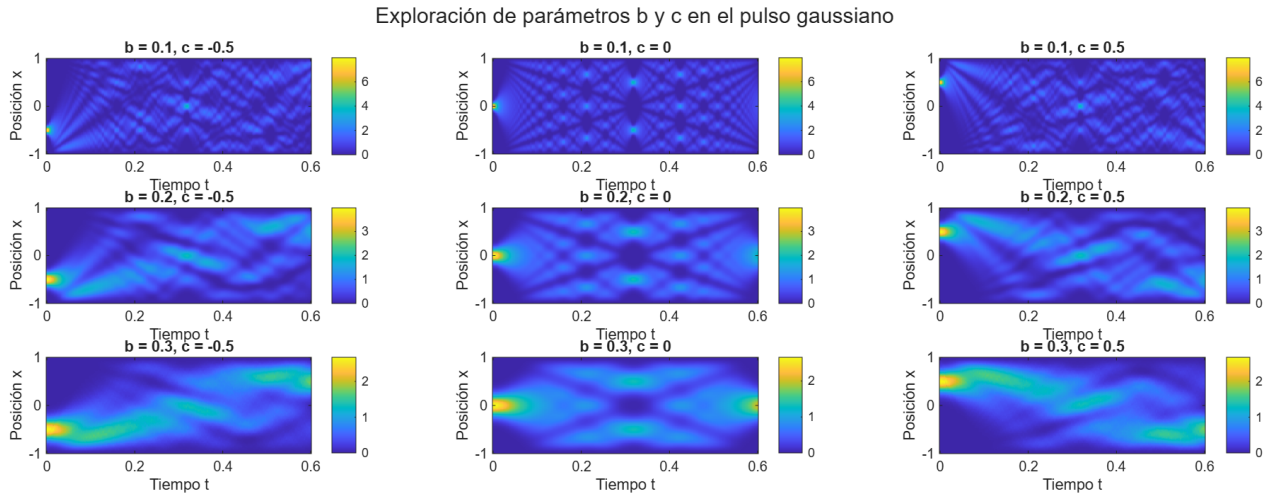
Expected position $\langle x \rangle(t)$.

$$\langle x \rangle(t) = \int x |\psi(x, t)|^2 dx,$$

evaluated numerically via the trapezoidal rule.


 Figure 4: Expected position $\langle x \rangle(t)$ for a Gaussian pulse.

The expectation value oscillates non-periodically, exhibiting three main peaks. These correspond to moments when constructive interference concentrates the probability density near the initial position and its mirror image, shifted slightly by dispersion and reflection effects.


 Figure 5: Probability density for different values of b and c .

Exploring parameters b and c . The parameter c controls the initial location of the pulse: a negative value shifts the particle toward the left side of the well, and vice versa. The parameter b governs the pulse width: small values produce narrow, highly localized pulses that require many energy modes, leading to rich interference patterns; larger values

yield broader pulses with smoother, less oscillatory dynamics and reduced contribution from excited states.

2.3 Case c: Traveling Gaussian Pulse

Adding a nonzero initial velocity v_0 introduces an initial momentum. This is implemented by multiplying the Gaussian envelope by a traveling-wave phase factor e^{ik_0x} , where $k_0 = Mv_0/\hbar$. The initial wavefunction becomes

$$\phi(x, 0) = A \exp \left[-\frac{(x - c)^2}{b^2} \right] \exp \left(i \frac{Mv_0}{\hbar} x \right).$$

Unlike the stationary case, the probability density center now propagates across the well before reflecting off the walls.

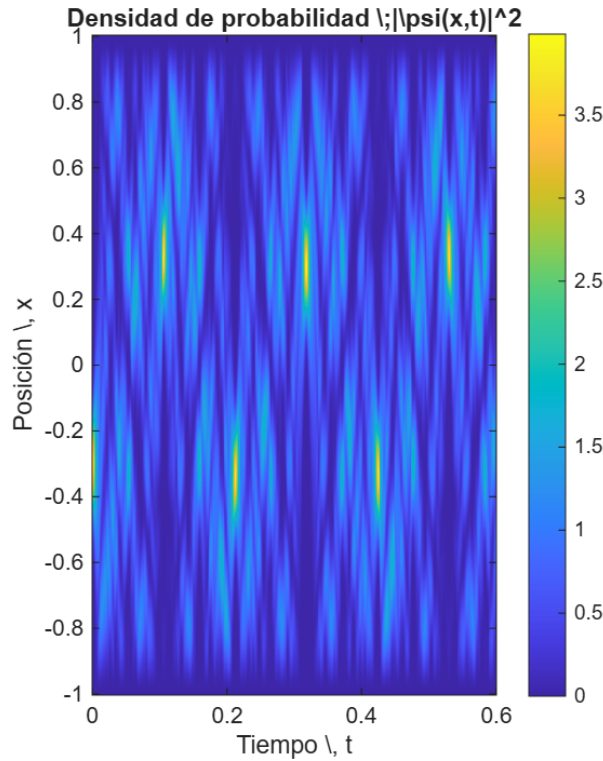


Figure 6: Probability density for a traveling Gaussian pulse ($v_0 = 4$).

Two-dimensional probability density $|\psi(x, t)|^2$. The displacement of the probability packet is clearly visible: the pulse propagates, bounces off the boundaries, and generates more pronounced interference patterns than the stationary case. Bright spots recurring at later times are consistent with partial revivals.

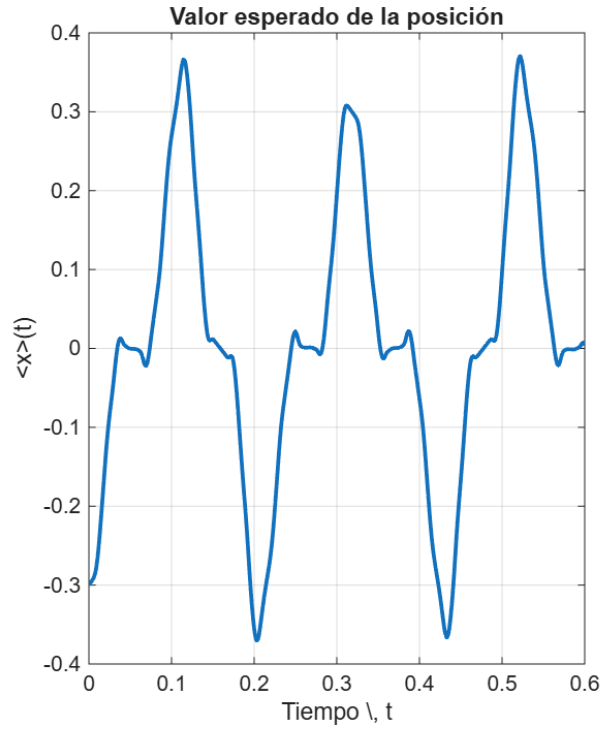
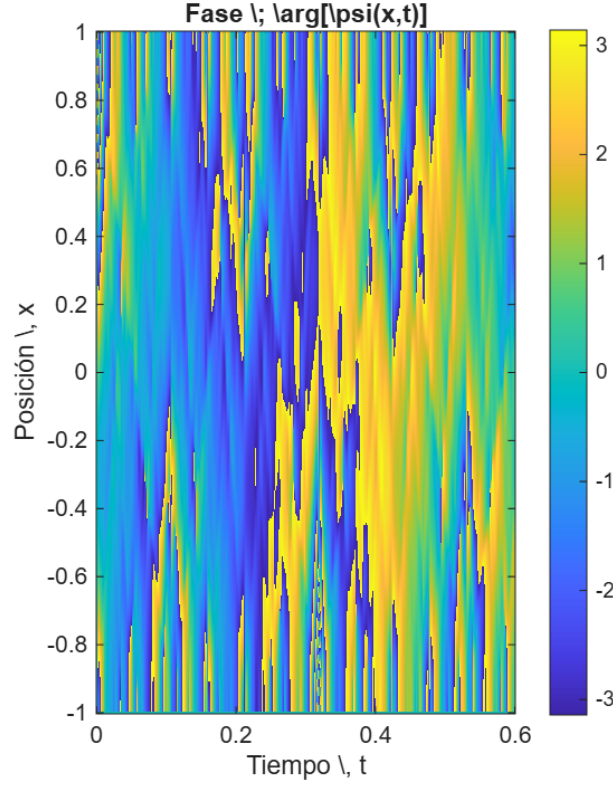


Figure 7: Expected position $\langle x \rangle(t)$ for the traveling Gaussian pulse.

Expected position $\langle x \rangle(t)$. The curve exhibits pronounced oscillations with the packet bouncing between the walls periodically, reflecting the initial momentum imparted to the state.

Figure 8: Phase $\arg[\psi(x, t)]$ of the wavefunction.

Phase portrait $\arg \psi(x, t)$. The phase portrait complements the probability density by revealing the full quantum dynamics of the state. The diagonal fringe patterns propagating from left to right confirm the presence of a nonzero initial momentum, and the sharp color discontinuities correspond to $\pi \rightarrow -\pi$ phase wrapping. Structures invisible in the density plot — such as the directional flow of phase fronts — become apparent here, making the phase portrait a powerful diagnostic tool.

Finite Potential Well: Dynamical Analysis

Eigenstate Normalization

1.1 Even States

The symmetric finite potential well is defined by

$$U(x) = \begin{cases} 0, & |x| < a \\ U_0, & |x| > a \end{cases}$$

The even eigenstates take the form

$$\psi_n(x) = \begin{cases} A_n \cos(k_n x), & |x| < a \\ B_n e^{-q_n |x|}, & |x| > a \end{cases}$$

with

$$k_n = \sqrt{\frac{2ME_n}{\hbar^2}}, \quad q_n = \sqrt{\frac{2M(U_0 - E_n)}{\hbar^2}}.$$

Continuity at $x = a$ requires

$$A_n \cos(k_n a) = B_n e^{-q_n a} \quad \Rightarrow \quad B_n = A_n \frac{\cos(k_n a)}{e^{-q_n a}}.$$

Applying the normalization condition and exploiting the symmetry of even states,

$$2 \left[\int_0^a |A_n|^2 \cos^2(k_n x) dx + \int_a^\infty |B_n|^2 e^{-2q_n x} dx \right] = 1.$$

Evaluating the integrals,

$$\int_0^a \cos^2(k_n x) dx = \frac{a}{2} + \frac{\sin(2k_n a)}{4k_n}, \quad \int_a^\infty e^{-2q_n x} dx = \frac{1}{2q_n} e^{-2q_n a},$$

and substituting B_n , we obtain

$$|A_n|^2 \left(a + \frac{\sin(2k_n a)}{2k_n} + \frac{\cos^2(k_n a)}{q_n} \right) = 1.$$

The normalization amplitudes are therefore

$$\boxed{A_n = \sqrt{\frac{q_n}{aq_n + 1}}, \quad B_n = A_n \frac{\cos(k_n a)}{e^{-q_n a}}}.$$

1.2 Odd States

The odd eigenstates are

$$\varphi_n(x) = \begin{cases} A_n \sin(k_n x), & |x| < a, \\ B_n \exp(-q_n x), & x > a. \end{cases}$$

Continuity of φ_n and φ'_n at $x = a$ gives

$$A_n \sin(k_n a) = B_n e^{-q_n a} \quad \Rightarrow \quad B_n = A_n \sin(k_n a) e^{q_n a},$$

$$A_n k_n \cos(k_n a) = -q_n B_n e^{-q_n a} \quad \Rightarrow \quad \tan(k_n a) = -\frac{k_n}{q_n}.$$

Applying normalization and using trigonometric identities with the odd-state characteristic equation,

$$\int_{-\infty}^{\infty} |\varphi_n(x)|^2 dx = A_n^2 \left(a - \frac{\sin(2k_n a)}{2k_n} + \frac{\sin^2(k_n a)}{q_n} \right) = A_n^2 \left(a + \frac{1}{q_n} \right) = 1,$$

yielding

$$A_n = \sqrt{\frac{q_n}{1 + aq_n}}, \quad B_n = \sqrt{\frac{q_n}{1 + aq_n}} \sin(k_n a) e^{q_n a}.$$

Eigenenergies and Eigenstates

The following reduced units are used throughout: $\hbar = 1$, $a = 1$, $M = \pi^2/8$, $U_0 = 110$.

Eigenenergies for $U_0 = 110$. The eigenenergies of the finite well are expressed in terms of the dimensionless parameter μ_n ,

$$E_n = \frac{\hbar^2 \pi^2}{2m(2a)^2} \mu_n^2 = V_0 \mu_n^2,$$

where V_0 is the ground-state energy of the infinite well. The allowed values μ_n are solutions to the characteristic equations

$$\textbf{Even:} \quad \tan\left(\frac{\pi}{2}\mu\right) = \frac{\sqrt{\mu_{\max}^2 - \mu^2}}{\mu}, \quad \textbf{Odd:} \quad \tan\left(\frac{\pi}{2}\mu\right) = -\frac{\mu}{\sqrt{\mu_{\max}^2 - \mu^2}},$$

with

$$\mu_{\max} = \sqrt{\frac{U_0}{V_0}}.$$

The total number of bound states is $N = \lceil \mu_{\max} \rceil = 11$. Solving the characteristic equations numerically gives the following allowed energies:

n	E_n
1	0.88869
2	3.5531
3	7.988
4	14.184
5	22.127
6	31.794
7	43.153
8	56.15
9	70.693
10	86.588
11	103.22

Table 1: Allowed energies for the finite well with $U_0 = 110$.

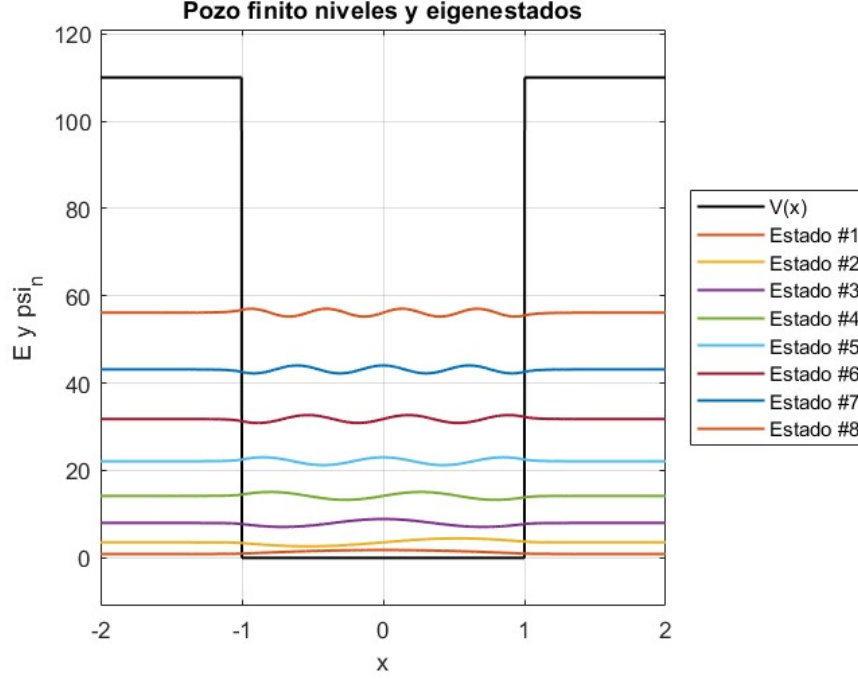


Figure 9: First eight eigenstates of the finite potential well.

First eight eigenstates. Each successive state carries higher energy than the previous one. Notably, higher-energy states exhibit greater penetration into the classically forbidden region beyond $|x| = a$ — a direct manifestation of quantum tunneling.

Time Evolution

3.1 Case a: Superposition of Four Eigenstates

The initial state is

$$\phi(x, 0) = A [\varphi_1(x) + 0.7\varphi_2(x) - 1.2i\varphi_3(x) + (2 - i)\varphi_4(x)],$$

with $A = 1/\sqrt{7.93}$ from normalization.

The time-evolved state is

$$\Psi(x, t) = \frac{1}{\sqrt{7.93}} [\varphi_1 e^{-iE_1 t/\hbar} + 0.7\varphi_2 e^{-iE_2 t/\hbar} - 1.2i\varphi_3 e^{-iE_3 t/\hbar} + (2 - i)\varphi_4 e^{-iE_4 t/\hbar}].$$

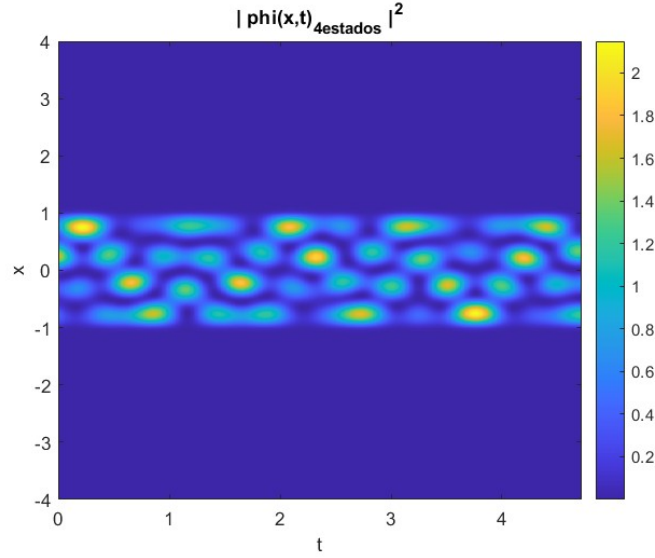


Figure 10: Time evolution of the probability density for a superposition of four eigenstates.

The heatmap encodes probability as a function of position (vertical axis) and time (horizontal axis). Warmer colors indicate higher probability. The interference patterns and amplitude modulations arise from the beating between the four incommensurable energy frequencies.

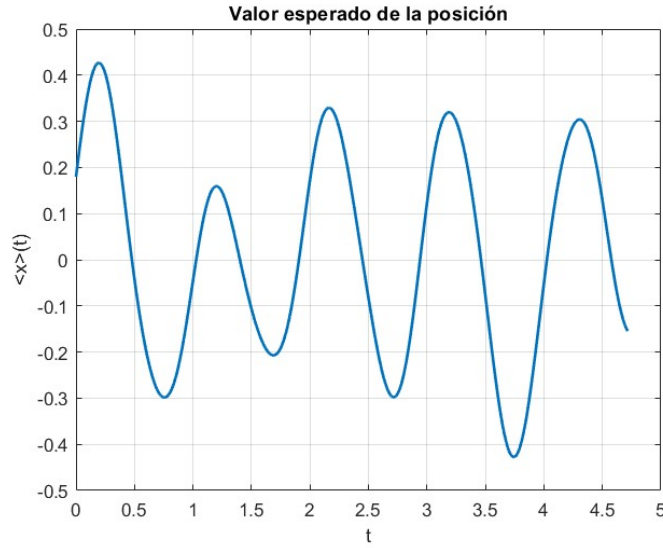


Figure 11: Expected position $\langle x \rangle(t)$ for the four-eigenstate superposition.

The oscillations in $\langle x \rangle(t)$ are not periodic in a simple sense, as they result from the combination of multiple frequency components simultaneously present in the state.

3.2 Case b: Gaussian Pulse

The Gaussian pulse is expanded in the finite well eigenstate basis:

$$c_n = \langle \phi_n | \phi(x, 0) \rangle, \quad \phi(x, 0) = 1.996 e^{-\frac{(x+0.3)^2}{0.2^2}}.$$

With 11 bound states available, the time-evolved state is

$$\Psi(x, t) = \sum_{n=1}^{11} \langle \phi_n | \phi(x, 0) \rangle \phi_n(x) e^{-iE_n t/\hbar}.$$

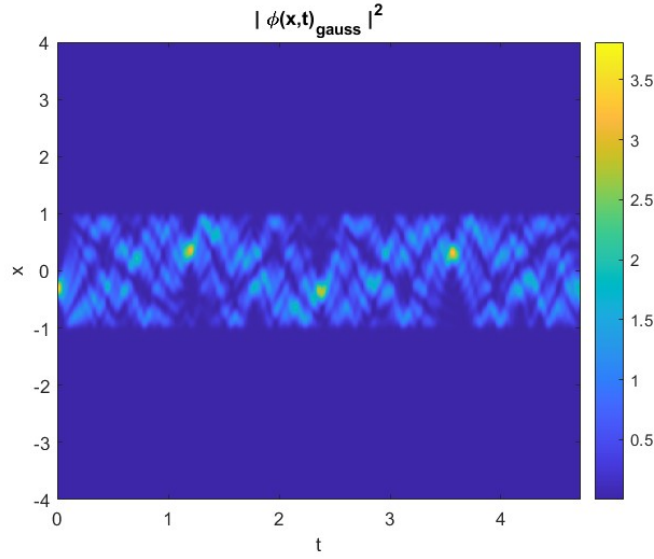


Figure 12: Probability density evolution for a Gaussian pulse in the finite well.

The heatmap shows how the initially localized Gaussian spreads and reflects inside the well. Warm regions trace the trajectory of the packet, and interference fringes become more complex over time due to multi-mode mixing.



Figure 13: Expected position $\langle x \rangle(t)$ for the Gaussian pulse.

The expectation value undergoes moderate oscillations, with amplitude variations arising from internal packet interference and boundary reflections.

3.3 Case c: Traveling Gaussian Pulse

Adding initial velocity $v_0 = 4$,

$$\phi(x, 0) = 1.996 e^{-\frac{(x+0.3)^2}{0.2^2}} e^{i \frac{M v_0}{\hbar} x}.$$

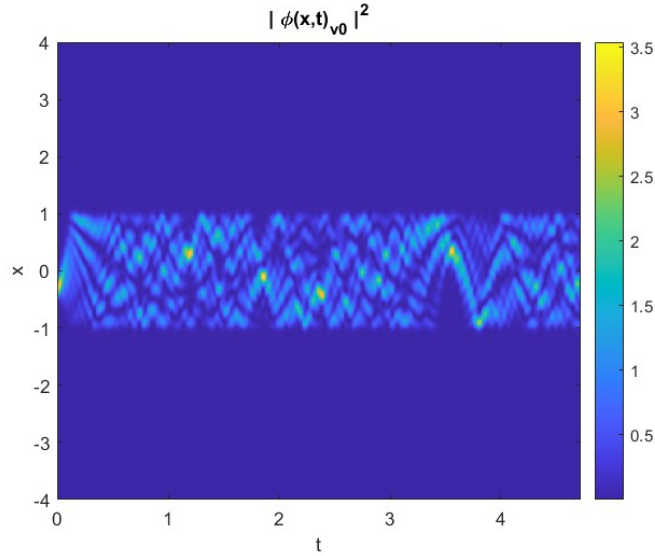


Figure 14: Probability density evolution for the traveling Gaussian pulse.

The probability packet now drifts, bounces off the walls, and generates more sharply defined interference structures than the stationary pulse. The bright spots recurring at later times are consistent with partial wave revivals.



Figure 15: Expected position $\langle x \rangle(t)$ for the traveling Gaussian pulse.

The curve exhibits more pronounced maxima and minima than the stationary case, reflecting the repeated impact of the packet against the well boundaries.

Animation. An animation of the probability density evolution for the traveling pulse is available at the following link: [Click here](#).

Discussion

The analyses of the infinite and finite potential wells reveal fundamental differences in quantum confinement, despite their structural similarity. In the infinite well, the spacing between consecutive energy levels scales as n^2 , a property that gives rise to exact periodic revivals of the wavefunction at integer multiples of $T = 4mL^2/\pi\hbar$. In the finite well, the allowed energies depend on the transcendental solutions μ_n^2 , which in turn depend on the well depth U_0 ; the level spacings are no longer integer multiples of a common frequency, and exact revivals do not occur in general.

The computational results clearly illustrate these differences. The same initial superposition of eigenstates — or the same Gaussian pulse — produces qualitatively distinct dynamical behaviors depending on whether it is placed in an infinite or finite well. Furthermore, adding an initial momentum to the Gaussian pulse introduces a directed propagation that is absent in the stationary cases, generating more structured interference patterns and sharper oscillations in the expected position.

Source Code

Infinite Well

Superposition of four eigenstates

```
1 % Time evolution: probability density and expected position
2 clear; clc;
3
4 %% Parameters
5 hbar = 1;
6 a = 1;
7 L = 2*a;
8 m = 1/8;
9 N = 4;
10 Trev = 4*m*L^2 / (pi*hbar); % Revival time
11
12 %% Spatial domain
13 Nx = 300;
14 x = linspace(-a, a, Nx);
15 dx = x(2) - x(1);
16
17 % Temporal domain
18 Nt = 400;
19 t = linspace(0, Trev, Nt);
20
21 %% Eigenfunctions and eigenenergies
22 phi = @(n, x) sqrt(1/a) * sin(n*pi*(x + a)/(2*a));
23 E = @(n) (n^2 * pi^2);
24
25 %% Complex coefficients
26 c = [1, 0.7, -1i*1.2, 2 - 1i];
27
28 %% Normalization constant
29 norma = sum(abs(c).^2); % = 7.93
30 A = 1/sqrt(norma);
31
32 %% Time evolution
33 psi_xt = zeros(Nx, Nt);
34 x_mean = zeros(1, Nt);
35
36 for ti = 1:Nt
37     psi_t = zeros(1, Nx);
38     for n = 1:N
39         phi_nx = phi(n, x);
40         psi_t = psi_t + c(n) * phi_nx .* exp(-1i * E(n) * t(ti)/
            hbar);
```

```

41     end
42     psi_t = A * psi_t;
43     psi_xt(:, ti) = abs(psi_t).^2;
44     x_mean(ti) = sum(conj(psi_t) .* x .* psi_t) * dx;
45 end
46
47 %% Plots
48 figure('Position', [100, 100, 1200, 500])
49
50 subplot(1,2,1)
51 imagesc(t, x, psi_xt)
52 xlabel('Time t'); ylabel('Position x')
53 title('Probability density |\psi(x,t)|^2')
54 set(gca, 'YDir', 'normal'); colorbar
55
56 subplot(1,2,2)
57 plot(t, x_mean, 'LineWidth', 2)
58 xlabel('Time t'); ylabel('<x>(t)')
59 title('Expected position'); grid on

```

Gaussian pulse

```

1 % Time evolution: Gaussian pulse
2 clear; clc;
3
4 %% Parameters
5 hbar = 1;
6 a = 1;
7 m = 1/8;
8 b = 0.2; % Pulse width
9 c0 = -0.3; % Initial center
10 Trev = 4 * m * (2*a)^2 / (pi * hbar);
11
12 %% Spatial and temporal domains
13 Nx = 300;
14 x = linspace(-a, a, Nx);
15 dx = x(2) - x(1);
16 Nt = 400;
17 t = linspace(0, Trev, Nt);
18
19 %% Normalization constant
20 A = (2/(pi * b^2))^(1/4);
21
22 %% Eigenstate basis
23 phi = @(n, x) sqrt(1/a) * sin(n*pi*(x + a)/(2*a));
24 E = @(n) (n^2 * pi^2 * hbar^2) / (2*m*(2*a)^2);
25 Nmax = 40;

```

```
26
27 %% Projection coefficients
28 gauss_prefac = exp(-((x - c0).^2) / b^2);
29 c = zeros(1, Nmax);
30 for n = 1:Nmax
31     c(n) = trapz(x, phi(n,x) .* (A * gauss_prefac));
32 end
33
34 %% Time evolution
35 psi_xt = zeros(Nx, Nt);
36 x_mean = zeros(1, Nt);
37 for ti = 1:Nt
38     psi_t = zeros(1, Nx);
39     for n = 1:Nmax
40         psi_t = psi_t + c(n) * phi(n,x) .* exp(-1i*E(n)*t(ti)/hbar);
41     end
42     psi_xt(:, ti) = abs(psi_t).^2;
43     x_mean(ti) = real(sum(conj(psi_t) .* x .* psi_t) * dx);
44 end
```

Traveling Gaussian pulse

```
1 % Traveling Gaussian pulse in an infinite well
2 clear; clc;
3
4 %% Parameters
5 hbar = 1; a = 1; m = 1/8;
6 b = 0.2; c0 = -0.3; v0 = 1.5;
7 k0 = m * v0 / hbar;
8
9 %% Domains
10 Nx = 300;
11 x = linspace(-a, a, Nx); dx = x(2)-x(1);
12 Nt = 400;
13 t = linspace(0, 0.6, Nt);
14
15 %% Initial wavefunction with momentum
16 psi0 = exp(-((x-c0).^2)/b^2) .* exp(1i*k0*x);
17 psi0 = psi0 / sqrt(trapz(x, abs(psi0).^2));
18
19 %% Eigenstate basis
20 Nmax = 80;
21 phi = @(n, xv) sqrt(1/a) * sin(n*pi*(xv+a)/(2*a));
22 E = @(n) (n^2*pi^2*hbar^2) / (2*m*(2*a)^2);
23
24 c = zeros(1, Nmax);
25 for n = 1:Nmax
```

```
26     c(n) = trapz(x, conj(phi(n,x)) .* psi0);
27 end
28
29 %% Time evolution
30 Psi_xt = zeros(Nx, Nt);
31 Theta_xt = zeros(Nx, Nt);
32 for ti = 1:Nt
33     psi_t = zeros(1, Nx);
34     for n = 1:Nmax
35         psi_t = psi_t + c(n)*phi(n,x).*exp(-1i*E(n)*t(ti)/hbar);
36     end
37     Psi_xt(:,ti) = psi_t;
38     Theta_xt(:,ti) = angle(psi_t);
39 end
40
41 rho_xt = abs(Psi_xt).^2;
42 x_mean = zeros(1,Nt);
43 for ti = 1:Nt
44     psi_t = Psi_xt(:,ti).';
45     x_mean(ti) = real(sum(conj(psi_t).*x.*psi_t)*dx);
46 end
```

Finite Well

Bound state computation and eigenstate superposition

```
1 clear; clc; close all;
2
3 %% Physical parameters
4 hbar = 1; a = 1; masa = pi^2/8; U0 = 110;
5
6 V0 = (hbar*pi)^2 / (2*masa*(2*a)^2);
7 mu_max = sqrt(U0 / V0);
8 K = sqrt(2*masa*U0) / hbar;
9
10 %% Solve for mu values (even and odd states)
11 [mu_par, mu_impar] = encontrarmu(mu_max);
12
13 k_par = (pi/(2*a)) * mu_par;
14 k_impar = (pi/(2*a)) * mu_impar;
15 q_par = sqrt(K^2 - k_par.^2);
16 q_impar = sqrt(K^2 - k_impar.^2);
17
18 E_par = (hbar^2 * k_par.^2) / (2*masa);
19 E_impar = (hbar^2 * k_impar.^2) / (2*masa);
20
21 Estados = struct('E',{}, 'es_par',{}, 'k',{}, 'q',{});
```

```

22 for i=1:numel(k_par)
23     Estados(end+1) = struct('E',E_par(i),'es_par',true, 'k',k_par(i)
        , 'q',q_par(i));
24 end
25 for i=1:numel(k_impar)
26     Estados(end+1) = struct('E',E_impar(i),'es_par',false, 'k',
        k_impar(i), 'q',q_impar(i));
27 end
28 [~,ord] = sort([Estados.E]); Estados = Estados(ord);
29
30 %% Eigenstate construction
31 Lcaja = 4*a;
32 x      = -(Lcaja:2*Lcaja/4000:Lcaja).';
33 idxC   = abs(x) <= a;
34 idxF   = ~idxC;
35 Nmax   = min(11, numel(Estados));
36 PSI    = zeros(numel(x), Nmax);
37 E_lista = zeros(Nmax,1);
38
39 for n = 1:Nmax
40     k = Estados(n).k; q = Estados(n).q; es_par = Estados(n).es_par;
41     A = sqrt(q/(1 + a*q));
42     C = A * (es_par*cos(k*a) + (~es_par)*sin(k*a));
43     psi = zeros(size(x));
44     if es_par
45         psi(idxC) = A * cos(k*x(idxC));
46         psi(idxF) = C * exp(-q*(abs(x(idxF))-a));
47     else
48         psi(idxC) = A * sin(k*x(idxC));
49         psi(idxF) = sign(x(idxF)) .* C .* exp(-q*(abs(x(idxF))-a));
50     end
51     PSI(:,n) = psi;
52     E_lista(n) = Estados(n).E;
53 end
54
55 %% Four-eigenstate superposition
56 coef_crudos = [1; 0.7; -1.2i; (2-1i)];
57 phi_ini     = PSI(:,1:4) * coef_crudos;
58 A0          = 1/sqrt(trapz(x, abs(phi_ini).^2));
59 coef_4      = A0 * coef_crudos;
60
61 [t3, phi_xt3, dens3] = evolucion_desde_coef(x, PSI(:,1:4), E_lista
    (1:4), hbar, Tfin, Nt, coef_4);
62
63 %% Helper functions
64 function [t, phi_xt, dens] = evolucion_desde_coef(x, PSI, E_lista,
    hbar, Tfin, Nt, c)
65     x = x(:);

```

```

66     t = linspace(0, Tfin, Nt);
67     fases = exp(-1i*(E_lista(:)/hbar)*t);
68     phi_xt = PSI * (c(:) .* fases);
69     dens = abs(phi_xt).^2;
70 end
71
72 function [mu_par, mu_impar] = encontrarmu(mu_max)
73     rhs_par = @(mu) sqrt(max(mu_max^2-mu.^2,0)) ./ max(mu,eps);
74     rhs_impar = @(mu) -max(mu,eps) ./ sqrt(max(mu_max^2-mu.^2,0));
75     f_par = @(mu) tan(pi*mu/2) - rhs_par(mu);
76     f_impar = @(mu) tan(pi*mu/2) - rhs_impar(mu);
77     tol = 1e-9;
78     opt = optimset('TolX',1e-12,'TolFun',1e-12,'Display','off');
79     mu_par = []; mu_impar = [];
80     mmax = floor((mu_max - tol)/2);
81     for m = 0:mmax
82         L = 2*m; R = min(2*m+1, mu_max-tol);
83         if R > L+tol && isfinite(f_par(L+tol)) && isfinite(f_par(R-
            tol))
84             try
85                 r = fzero(f_par, [L+tol, R-tol], opt);
86                 if r>L && r<R, mu_par(end+1,1) = r; end
87             catch; end
88         end
89         L = 2*m+1; R = min(2*m+2, mu_max-tol);
90         if R > L+tol && isfinite(f_impar(L+tol)) && isfinite(f_impar
            (R-tol))
91             try
92                 r = fzero(f_impar, [L+tol, R-tol], opt);
93                 if r>L && r<R, mu_impar(end+1,1) = r; end
94             catch; end
95         end
96     end
97     mu_par = unique(sort(mu_par));
98     mu_impar = unique(sort(mu_impar));
99 end

```